

Physics Reading Plan

AI-assisted workflows for textbooks and papers — Parker Lewis

Core principle

Reading and AI-consulting are separate phases. AI lives at the boundaries of reading — orientation before, debug and synthesis after — not inside it. The slow, hand-worked pass is where understanding actually forms and cannot be skipped.

Part 1 — Textbook chapter workflow

Per-section loop. Repeat for each section of the chapter, then do the chapter-level closing steps.

Step 1 — Fast read & mark questions

- Read the section straight through at pace — do not stop for confusing spots
- Mark every question in a running scratch doc as you go
- Tag each question by type so the distribution tells you something:
 - (a) notation / definition
 - (b) derivation step
 - (c) physical intuition
 - (d) connection to other material
- If the question is “what does this symbol mean” or “why this step” — skim forward 2 pages before writing it down; the author may resolve it
- If the question is “what’s the physical motivation” or “how does this connect to X” — write it down, the book probably won’t tell you

Step 2 — AI LaTeX synthesis

- Paste the section (or describe it clearly) and request a textbook-level LaTeX writeup
- The writeup should include:
 - physical motivation before and after each result
 - full derivation with every step shown — more steps than the book, not fewer
 - worked mini-examples where useful
 - boxed / color-coded key results
 - explicit flags on subtle points or places the author skipped steps
 - 2–3 conceptual check questions at the end
- Specify notation conventions up front (metric signature, natural units, index placement) to match the book
- Keep all writeups in one growing .tex file per textbook — builds a personal companion text over time

*The synthesis is **not** a weaker substitute for the textbook. It is a different artifact — more pedagogical, more verbose, tuned to you — that sits alongside the book. The book is ground truth; the writeup is the teacher who walks you through it.*

Between steps 2 and 3 — self-check

- Close the book and close the synthesis
- Write 3–4 sentences from memory: what did this section prove?
- Compare your version to the synthesis — the gap shows exactly what didn't stick

Step 3 — Slow re-read (always, no exceptions)

- Re-read the section from the textbook, slower, with paper and pencil
- Work every derivation by hand — do not skip
- Do **not** flip back to the AI synthesis during this pass — it becomes a crutch
- Budget: if skim took 20 min and synthesis read took 15 min, slow pass should be 45–90 min
- This is where recognition becomes reconstruction — the difference between “I’ve seen this” and “I can do this”

The trap to watch

After a clean synthesis, the slow read will feel redundant. That is the moment the workflow is being tested. Push through every time. Skipping here is the single failure mode that collapses the whole plan.

Chapter-level close (after all sections done)

- Output the full question doc as a LaTeX file
- Before opening any problems, try to answer your earlier questions from memory
- Problem selection:
 - target problems at the fuzzy answers, not the clean ones
 - 6–8 well-chosen problems > grinding through all 20
- Final 10-min AI pass: paste your question doc with your answers, ask for critique and for any important question you didn't ask

Textbook loop — quick reference

Phase	Action	AI role
1. Fast read	Skim section, mark & tag questions	None
2. Synthesis	AI writes textbook-level LaTeX of section	Author
Check	Write 3–4 sentences from memory	None
3. Slow read	Re-read from book, derivations by hand	None (resist!)
Chapter end	Answer own questions, then problems	Critique at end

Part 2 — Physics paper workflow

Papers announce; textbooks teach. Don't read them linearly — even senior people don't. Structure is optimized for archival reference, not learning.

Before you start

- Open a scratch doc alongside the paper for definitions, acronyms, one-line claims
- Decide up front: triage (is this worth reading?) or real read?

Pass 1 — Orient (10 min, no AI)

- Read: title, abstract, intro's first and last paragraphs, figure captions, conclusion
- Answer four questions:
 - what problem?
 - what's new?
 - what did they measure / compute?
 - does it matter to me?
- If answer to the last is no — stop. Papers are cheap to abandon.

Pass 2 — AI context dump

- Paste abstract + intro to AI
- Ask for:
 - current state of the subfield
 - what the authors are reacting against
 - 3–5 key prior papers
 - notation and jargon to expect
- Skim abstracts of the 2–3 most-cited prior papers before continuing

Pass 3 — Figures first

- Go through every figure in order
- Read only the caption and the paragraph in the body discussing each figure
- Ask AI to explain any figure you can't parse: what's plotted, the takeaway, what a null result would look like
- Understanding the figures = understanding 70% of the paper

Pass 4 — Body, strategically

- Read methods and results; **skip derivations** on first pass — mark them
- Goal: understand *what was done*, not yet *how*
- Mark unclear equations, unfamiliar techniques, unmotivated claims
- For experimental papers: study the systematic uncertainty table — it shows what they actually worried about
- For theory papers: read the conclusion before the body — theorists bury the punchline

Pass 5 — AI debug session

- Batch **all** marked questions in one go
- Terse derivation: paste the equation, ask for missing steps, “in the style of a textbook”
- Unfamiliar technique: ask for 2-paragraph primer + standard reference
- Unmotivated claim: ask “why is this approach natural here?”

Pass 6 — Honest test

- Close the paper. Write 3–5 sentences:
 - - what they did
 - what they found
 - why it matters
 - what you'd do next
- If you can't, you haven't read it yet — go back to whichever pass you shortchanged

Anti-fatigue rules (apply throughout)

- 25-minute blocks, then a real break (walk the dog, not scroll the phone)
- Paper-reading burns ~3× the fuel of textbook-reading — treat it that way
- After each section, write one sentence: “this section established X”
- If foggy: paste the read-so-far to AI, ask for a running summary to rebuild the mental model
- If you re-read the same paragraph twice without it landing — stop. Buffer is full.
- Stop when tired, not when done. Pushing through fatigue creates false memories of having read things.
- For long / complex papers: spend 5 min drawing a visual map — boxes for sections, arrows for how they feed each other. Keep it next to you.

Why you lose the thread

The losing-context feeling is almost always losing the macro structure while focused on micro detail. When it happens, the fix is usually three paragraphs back — chase the weak link now, not push forward.

After the paper (for work close to dissertation)

- Rebuild a key plot from their description, or code up their method on toy data
- Add new jargon to a running personal glossary
- File the paper with a one-paragraph summary in your own words

Per-session checklist

- Scratch doc open
- 25-min timer running
- One sentence written per section
- Visual map nearby for long papers
- AI used for context and debug, **not** live translation

Calibrate expectations

A good paper at your level takes 3–8 hours to really read, spread over multiple sittings. “I read it but I don’t get it” after 45 minutes isn’t failure — it’s the normal state after pass 3. The “aha” often hits a day later while doing something else entirely.